

James Petley

Curriculum Vitae



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Research Interests

I have spent the last four years building the appropriate skills to investigate, and a deep interest in understanding, galaxy formation, star formation history and formation of structure on a larger scale through a combination of simulation and observation. I believe I am well suited to research and would like to be able to push myself, particularly with new data from various surveys scheduled to be releasing very soon.

It is my wish to advance my knowledge with an expert supervisor at a prestigious university while completing a PhD. I am exceptionally hard working and like to take pride in my work, a research setting would be perfect for me.

Research Experience

Investigating the Most Luminous Starbursts in the Universe, *Durham University*, 2019–2020
Supervisors - Prof. Ian Smail and Prof. Mark Swinbank, Grade - Ongoing.

This project combines data from several galaxy surveys (primarily ALMA's AS2COSMOS sample) in an attempt to understand the galaxy formation processes and histories of very bright, high-redshift ($1 < z < 6$) and dusty galaxies. By matching galaxies across a large range of wavelengths and detectors, it is possible to determine many parameters of the galaxies. MAGPHYS FORTRAN code (da Cunha et al. 2008) can estimate these properties before comparison with the general population and work on the significance of these galaxies in relation to our local neighbourhood. This project requires extensive use of Python for data analysis and visualisation. It also requires understanding of how each galaxy survey was conducted and investigation into how to deal with non-detections, limits and issues with each of the surveys. A likely outcome for this project is a basic star-formation history of the universe.

How viable are public contributions to asteroid astronomy?, *Durham University*, Super- 2019
visors - Prof. Alastair Edge and Prof. Mark Swinbank, Grade - 73%.

This project used telescopes on the roof of the Durham Physics building to track several asteroids over two months. I used Python scripts to make calls to star catalogues and NASA's JPL HORIZONS system so that the photometry and identification of the asteroid in the image could be achieved automatically.

Jovian Planets: Friend or foe? (Python Project), *Durham University*, Supervisor - 2018–2019
Dr Nigel Metcalfe, Mark - 70%.

The project began with developing N-body simulation code in Python that could simulate a reduced version of the solar system (Sun, Earth and Jupiter). I then extended this to be a model of planetary formation from smaller planet-embryos. I investigated the effect that the presence of Jupiter and Saturn would have on the formation processes.

Dark Matter Distribution in NGC 3198, *Supervisor - Prof. Mark Swinbank and Dr Richard Wilman*, *Durham University*, Mark - 77%. 2018

This project used the telescopes at Durham to observe galaxy NGC 3198 and calculate various properties of the stars and dark matter halo.

Education

Physics and Astronomy, MPhys, *Durham University*, Durham, On course for a 2016–Current
1st Class Degree.

Baxter Prize from Hatfield College in 2017 and 2018 for Academic Achievement

Secondary School, *King Edward VI Aston School*, Birmingham, State Grammar School 2009–2016
in Central Birmingham.

Qualifications

A-Levels: 2016

- Further Mathematics = A*
- Mathematics = A*
- Physics = A

AS-Levels: 2015

- French = *B*

AS-Levels: 2014

- Philosophy and Ethics = *A*

GCSEs: 2014

- Mathematics, Physics, Chemistry, Biology, English Literature, English Language, French, Religious Studies = *A**
- History, Music = *A*

Programming and Computer skills

Python: Proficient in Python for Data Analysis and Physics Problem Solving. I also have some experience in web development with Django and Flask. I recently won a prize at Durham Hackathon for a Flask project created in 24 hours.

LATEX: I have written nearly all of my academic documents at University using LATEX and feel very comfortable doing so. I have also used it for different types of documents including journal-style papers, progress reports during projects and a poster.

C++: I have some understanding of the language but have not had the chance to use it in a practical or work setting.

Machine Learning: I have a small amount of machine learning experience mostly through experimentation with Google Keras and TensorFlow. This is an area I would like to learn a lot more about over the next few years.

Experience

Paid

Summer Internship, Drone Industries Ltd., Gateshead, 7 weeks.

2019

I won the chance to work for a company in the space or space technologies industry through a Santander scheme. I was mostly given jobs related to website design and structure but I managed to implement some extra features such as interactive maps combined with drone imagery using javascript. I also used OpenCV to help identify different types of plants in drone images for the local council. I combined artificial intelligence with archive images from a local museum to pull out useful text and key aspects of their extensive collection.

Tutor, Advantage 11+, Birmingham.

2016–Current

I have been tutoring children between the ages of 8 and 11 for several years. This has allowed me to:

- As a leader, manage a class of 25 young students and two other tutors
- Plan series of lessons between 2 and 4 weeks long
- Deliver short lessons on key topics in a clear and effective manner
- Communicate well with parents, giving advice on what their children need to work on and how they should be studying

Musician, Durham.

2016–Current

I have played in several bands while at Durham University. I play clarinet (Grade 8 distinction), saxophones, piano (grade 7 merit) and I am learning the flute. I have been able to:

- Negotiate prices and payment with large university organisations for balls and other events
- Lead a big band of 18 people as president. Manage several thousand pounds for a year as well as providing leadership and direction.
- Organise a tour for 16 people to travel to Amsterdam. Liaise with transport and accommodation providers, set up venues to perform abroad and budget based on income for the year.
- Record two albums with a band in Durham and release them online.

Volunteering

Teaching Assistant, Wilson Stuart School, Birmingham.

2014

I volunteered as a teaching assistant at this Special Academy for disabled students. I worked with a class of students of very different abilities and needs. I improved:

- My confidence in responding to stressful or unusual situations (e.g. student having a seizure)
- Communication skills with the students and also teachers. I even learnt some "makaton" sign language
- Teaching skills. How to teach a 14-year-old student practical life skills when they struggle to count past 10.

Experience Week, University of Birmingham, Birmingham, Physics and Astronomy Department. **2014**

A week of lectures and largely self-led experiments. Confirmed my interest in studying Physics and gave me the chance to learn rapidly about unseen equipment and skills in a practical manner.

Languages

French: Moderate

B at AS Level

Further Achievements

Duke of Edinburgh Award: I have completed my Gold Duke of Edinburgh award which involved many hours of volunteering, sporting activity and new experiences. Prince Edward presented me with the award in 2017.

Physics Olympiad: I achieved a Bronze in the Cambridge University Physics Olympiad for 2015, which put me in the top 400 students who took the test.

Running Despite Adversity: I was diagnosed with Juvenile Arthritis, a genetic disease which affects my joints, when I was 15. Despite this, I have been able to continue sporting activity and, in particular, running where I have been able to run half marathons (starting 2014) and recently a full marathon(2018). I also play football and ultimate frisbee most weeks.